

2nd Day Internship [8th June 2026] CoE - MathWorks DeepTech Labs

https://in.mathworks.com/help/matlab/index.html?s_tid=hc_panel

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```
RR_12=5
```

```
RR_12 = 5
```

```
clc=5
```

```
clc = 5
```

```
func=666
```

```
func = 666
```

```
namelengthmax
```

```
ans = 2048
```

```
exist aabb
```

```
ans = 0
```

```
exist display
```

```
ans = 5
```

```
% check to see if A is greater than B
A=10; B=20; if A > B "greater" elseif
A < B
"less"
end
```

```
ans = "less"
```

```
round(100*rand(1,10))
```

```
ans = 1x10
    71     3    28     5    10    82    69    32    95     3 true
or false
```

```
A = true(3,4)
```

```
A = 3x4 logical array
    1     1     1     1
    1     1     1     1
    1     1     1     1
```

```
false(4,6)
```

```
ans = 4x6 logical array    0
    0     0     0     0     0
    0     0     0     0     0     0
    0     0     0     0     0     0
    0     0     0     0     0
```

```
magic(2)
```

```
ans = 2x2
    1
    3
    4
    2
```

```
tm = magic(5)
```

```
tm = 5x5
    17    24     1
    8     15
    23     5     7    14    16
    4     6    13    20    22
   10 12    19    21    3 11 18
   25
    2     9
```

```
tm1=tm(1:2,1:2)
```

```
tm1 = 2x2
    17
   24
    23     5
```

```
tm2=tm(:,3:4)
```

```
tm2 = 5x2
    8
    7    14
   13    20
   19    21
   25     2
```

```
true(3,2,2)
```

```
ans = 3x2x2 logical array ans(:,:,1)  
=
```

```
    1    1  
    1    1    1  
1
```

```
ans(:,:,2) =
```

```
    1    1  
1    1  
1    1 ans  
= 1
```

greater logical array of selected

sparsity

```
A = logical(sparse(5,3)); whos
```

A

Name	Size	Bytes	Class	Attributes
A	5x3	41	logical	sparse

```
F = false(4,like=A); whos
```

F

Name	Size	Bytes	Class	Attributes
F	4x4	49	logical	sparse create

grids

```
y = linspace(0,100)
```

```
y = 1x100
```

```
    0    1.0101    2.0202    3.0303    4.0404    5.0505    6.0606    7.0707 ...
```

```
y = linspace(0,10,100)
```

```
y = 1x100
```

```
    0    0.1010    0.2020    0.3030    0.4040    0.5051    0.6061    0.7071 ...
```

log space

```
y = logspace(5,4)
```

```
y = 1×50
5
10 x
```

```
1.0000    0.9541    0.9103    0.8685    0.8286    0.7906    0.7543    0.7197 ...
```

```
y = logspace(5,4,100)
```

```
y = 1×100
5
10 x
```

```
1.0000    0.9770    0.9545    0.9326    0.9112    0.8902    0.8697    0.8498 ...
```

```
y = logspace(5,3.14)
```

```
y = 1×50
5
10 x
```

```
1.0000    0.9163    0.8396    0.7693    0.7050    0.6460    0.5919    0.5424 ...
```

```
Y = logspace(5,3.14,100)
```

```
Y = 1×100
5
10 x
```

```
1.0000    0.9577    0.9171    0.8783    0.8411    0.8055    0.7714    0.7387 ...
```

Frequency space

```
[f1,f2] = freqspace(5)
```

```
f1 = 1×5
-0.8000    -0.4000         0     0.4000     0.8000 f2
= 1×5
-0.8000    -0.4000         0     0.4000     0.8000
```

```
[f1,f2] = freqspace([0 5])
```

```
f1 = 1×5
-0.8000    -0.4000         0     0.4000     0.8000 f2 = 1×0 empty double row vector mesh grid 2D
```

```
x = 1:3; y
```

```
= 1:5;
[X,Y] = meshgrid(x,y)
```

```
X = 5×3
1     2     3
1     2     3
1     2     3
1     2     3
1     2     3
```

```
Y = 5×3
```

1 1

2

2 2

2

3 3

7

4 4

7

5 5

7

MESH GRID 3D

```
x = 0:2:6; y
= 0:1:6; z =
0:3:6;
[X,Y,Z] = meshgrid(x,y,z);
```

size of array

```
aa=magic(7)
```

aa = 7×7

30	39	48	1	10	19	28
38	47	7	9	18	27	29
46	6	8	17	26	35	37
5	14	16	25	34	36	45
13	15	24	33	42	44	4

21	23	32	41	43	3	12
----	----	----	----	----	---	----

22	31	40	49	2	11	20
----	----	----	----	---	----	----

 number of elements in 3D matrix

```
A = magic(4);
```

```
A(:,:,2) = A'
```

A =

A(:,:,1) =

16	2	3	13
5	11	10	8
9	7	6	12
4	14	15	1

A(:,:,2) =

16	5	9	4
2	11	7	14
3	10	6	15
13	8	12	1

determine scalar and vector from

matrix

```
A = [1 2; 3 4]; TF  
= isscalar(A)
```

```
TF = logical  
0
```

```
A = [1 2; 3 4];  
TF = isvector(A)
```

```
TF = logical  
0
```

is matrix

```
A3 = zeros(1,3,2); TF  
= ismatrix(A3)
```

```
TF = logical  
0 is row and  
column V =  
rand(5,1); tf =  
isrow(V)
```

```
tf = logical  
0
```

```
V = rand(5,1); tf =  
iscolumn(V)
```

```
tf = logical  
1
```

resize

```
aa=magic(16)
```

```
aa = 16×16
```

```
    256     2     3    253    252     6     7    249    248    10    11    245    244 ...  
    17    239    238     20     21    235    234     24     25    231    230     28     29  
    33    223    222     36     37    219    218     40     41    215    214     44     45  
208 50     51    205    204     54     55    201    200     58     59    197    196  
    192     66     67    189    188     70     71    185    184     74     75    181    180  
     81    175    174     84     85    171    170     88     89    167    166     92     93  
     97    159    158    100    101    155    154    104    105    151    150    108    109  
144 114    115    141    140    118    119    137    136    122    123    133    132  
    128    130    131    125    124    134    135    121    120    138    139    117    116  
145 111    110    148    149    107    106    152    153    103    102    156    157  
    161     95     94    164    165     91     90    168    169     87     86    172    173  
    80    178    179     77     76    182    183     73     72    186    187     69     68
```

```

        64   194   195    61    60   198   199    57    56   202   203    53    52
209 47    46   212   213    43    42   216   217    39    38   220   221
      225    31    30   228   229    27    26   232   233    23    22   236   237
      ⋮

```

```
head(aa',2)'
```

```
ans = 16×2
```

```

    256     2
     17    239
     33    223
    208     50
    192     66
81    175
     97    159
144 114
    128    130
145 111    161    95
     80    178
     64    194
    209     47
    225     31
     ⋮

```

resize vector

```
A = [1;3;5;7]; B1
= resize(A,2)
```

```
B1 = 2×1
```

```
1
```

```
3
```

Reshaping

```
A = 1:10;
B = reshape(A,[5,2])
```

```
B = 5×2
```

```

1      2
2      2
3      7
4      9
5      9

```

permute

```
rng default A = rand(3,4,2)
```

```
A =  
A(:,:,1) =  
    0.8147    0.9134    0.2785    0.9649  
    0.9058    0.6324    0.5469    0.1576  
    0.1270    0.0975    0.9575    0.9706
```

```
A(:,:,2) =  
    0.9572    0.1419    0.7922    0.0357  
    0.4854    0.4218    0.9595    0.8491  
    0.8003    0.9157    0.6557    0.9340
```

```
B = permute(A,[3 2 1])
```

```
B =  
B(:,:,1) =  
    0.8147    0.9134    0.2785    0.9649  
    0.9572    0.1419    0.7922    0.0357
```

```
B(:,:,2) =  
    0.9058    0.6324    0.5469    0.1576  
    0.4854    0.4218    0.9595    0.8491
```

```
B(:,:,3) =  
    0.1270    0.0975    0.9575    0.9706
```

```
0.8003    0.9157    0.6557    0.9340 rearranging
```

```
aa = magic(4)
```

```
aa = 4x4    16    2    3  
    13  
    5    11    10    8  
    9    7    6    12 4  
    14    15    1
```

```
sort(aa)
```

```
ans = 4x4    4    2  
    3    1  
5    7    6    8  
    9    11    10    12  
    16    14    15    13
```

```
A = magic(4)
```

```
A = 4x4    16    2    3    13  
    5    11    10    8  
    9    7    6    12
```

4 14 15 1

```
B = A.'
```

B = 4×4

	16	5	9	4
2	11	7	14	
3	10	6	15	
	13	8	12	1

c Transpose

```
A = [2 1; 9 7; 2 8; 3 5]
```

A = 4×2

	2	1
	9	7
2	8	
3	5	

```
B = A'
```

B = 2×4

2	9	2	3
1	7	8	5

circ shift

```
A = (1:10)'
```

A = 10×1

1
2
3
4
5
6
7
8
9
10

```
Y = circshift(A,3)
```

Y = 10×1

8
9
10
1
2
3
4
5
6

7 uint

32

```
y = uint32(10); whos
```

y

Name	Size	Bytes	Class	Attributes	y
1x1	4	uint32		double precision	

arrays

```
x = 10; whos x
```

Name	Size	Bytes	Class	Attributes	x
1x1	8	double		is	

NAN

```
A = 0./[-2 -1 0 1 2]
```

```
A = 1x5
    0    0 NaN    0    0
```

```
TF = isnan(A)
```

```
TF = 1x5 logical array
    0    0    1    0    0
```

```
str = "i want you take i want"
```

```
str = "i want you take i want"
```

```
str=split(str)
```

```
str = 6x1 string "i"
"want"
"you"
"take"
"i"
"want"
```

```
str = ["Thanu","Adi"; "tharun","swathi"; "cherry","sharmila"]
```

```
str = 3x2 string
"Thanu"    "Adi"
"tharun"   "swathi"
"cherry"   "sharmila"
```

```
str = join(str)
```

```
str = 3x1 string "Thanu
Adi"
"tharun swathi"
"cherry shar..."
```

```
s1 = ["Adi is very intelligent boy", "he is", "LT509"]
```

```
s1 = 1x3 string
"Adi is very intelligent... "he is" "LT509"
```

```
p1 = "Adi"
```

```
p1 = "Adi"
```

```
contains(s1,p1)
```

```
ans = 1x3 logical array    1
    0    0
```

```
DateStrings = {'2014-05-26'; '2014-08-03'}; t = datetime(DateStrings, 'InputFormat', 'yyyy-MMdd')
```

```
t = 2x1 datetime 26-May2014
    03-Aug-2014
```

Data strings

```
DateStrings =
{'2014-05-26T13:30-05:00'; '2014-08-26T13:30-04:00'; '2014-09- 26T13:30Z'}
```

```
DateStrings = 3x1 cell
'2014-05-26T13:30-05:00'
'2014-08-26T13:30-04:00'
'2014-09-26T13:30Z'      convert
```

excel data into date

```
X = [39558, 39600; 39700, 39800]
```

```
X = 2x2
    39558    39600
    39700    39800
```

```
t = datetime(X, 'ConvertFrom', 'excel')
```

```
t = 2x2 datetime
20-Apr-2008  01-Jun-2008
09-Sep-2008 18-Dec-2008
```

Structures

```
data.x = linspace(0,2*pi);
data.y = sin(data.x); data.title
= 'Y = sin(x)'
```

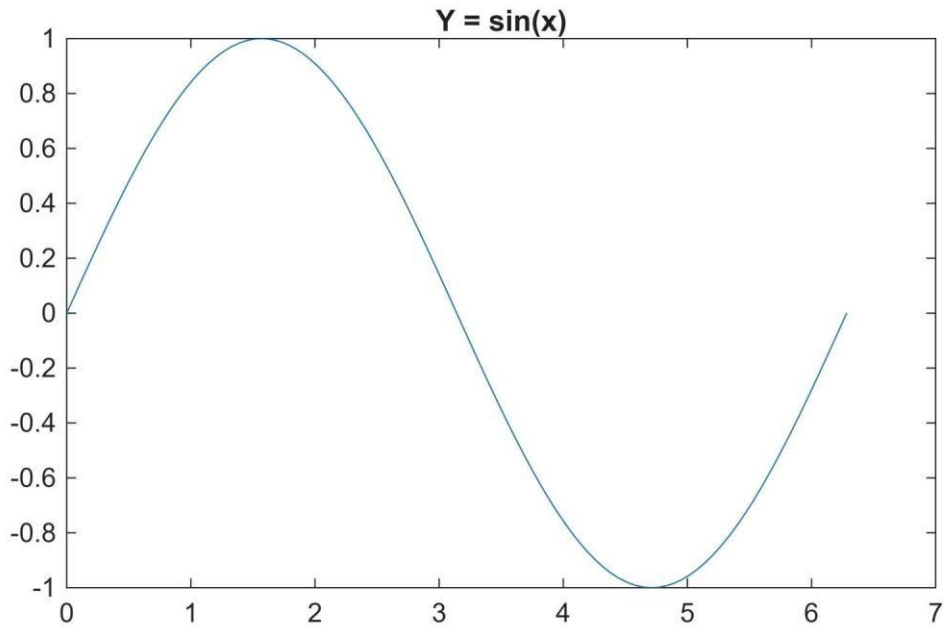
```
data = struct with fields:      x: [0 0.0635 0.1269 0.1904 0.2539 0.3173 0.3808 0.4443 0.5077 0.5712 0.6347
0.6981 0.7616 0.8251 0.8885 0.95      y: [0 0.0634 0.1266 0.1893 0.2511 0.3120 0.3717 0.4298 0.4862 0.5406 0.5929
0.6428 0.6901 0.7346 0.7761 0.81      title: 'Y = sin(x)']
```

```
zz = data.x
```

```
zz = 1×100
```

```
0    0.0635    0.1269    0.1904    0.2539    0.3173    0.3808    0.4443 ...
```

```
plot(data.x,data.y) title(data.title)
```



```
field = 'f'
```

```
field = 'f'
```

```
value = {'some text';
[10, 20, 30];      magic(5)}; s = struct(field,value)
```

```
s = 3×1 struct
```

Fields	f
1	'some text'
2	[10,20,30]
3	5×5 double

structures with multiple fields

```
field1 = 'f1'; value1 = zeros(1,10); field2 = 'f2'; value2 = {'a',
'b'}; field3 = 'f3'; value3 = {pi, pi.^2}; field4 = 'f4'; value4
```

```
= {'fourth'}; s = struct(field1,value1,field2,value2,field3,value3,field4,value4)
```

```
s = 1x2 struct
```

Fields	f1	f2	f3	f4
1	[0,0,0,0,0,0,0,0,0]	'a'	3.1416	'fourth'
Fields	f1	f2	f3	f4
2	[0,0,0,0,0,0,0,0,0]	'b'	9.8696	'fourth'

```
s(1)
```

```
ans = struct with fields:
f1: [0 0 0 0 0 0 0 0 0]
f2: 'a'
f3: 3.1416
f4: 'fourth'
```

```
s(2)
```

```
ans = struct with fields:    f1:
[0 0 0 0 0 0 0 0 0]    f2:
'b'    f3: 9.8696    f4: 'fourth'
```

Functions

```
fun = 'round'; x1 =
pi; y = feval(fun,x1)
```

```
y = 3
```

```
products = ["Tomato" "carrot" "Mango" "Mushroom"]
```

```
products = 1x4 string
"Tomato"    "carrot"    "Mango"    "Mushroom"
```

```
Prices = [1 0.5 2.50 1.99]; d = dictionary(products,Prices)
```

```
d = dictionary (string ↦ double) with 4 entries:
```

```
"Tomato"    ↦ 1
"carrot"    ↦ 0.5000
"Mango"     ↦ 2.5000
"Mushroom"  ↦ 1.9900
```

Time series

```
ts1 = timeseries([5 10 15 20 25]','Name','MyTimeSeries')
```

timeseries

Common Properties:

 Name: 'MyTimeSeries'

 Time: [5x1 double]

 TimeInfo: [1x1 tsdata.timemetadata]

 Data: [5x1 double]

 DataInfo: [1x1 tsdata.datametadata] More

properties, Methods

```
ts2 = timeseries(rand(2,5))
```

timeseries

Common Properties:

 Name: 'unnamed'

 Time: [2x1 double]

 TimeInfo: [1x1 tsdata.timemetadata]

 Data: [2x5 double]

 DataInfo: [1x1 tsdata.datametadata]

More properties, Methods